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A comparison of two exact methods for passenger railway rolling stock (re)scheduling



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ABSTRACT

The assignment of rolling stock units to timetable services in passenger railways is an important optimization problem that has been addressed by many papers in different forms. Solution approaches have been proposed for different planning phases: strategic, tactical, operational, and real-time planning. In this paper we compare two approaches within the operational and real-time planning phase. The first exact approach is based on a known Mixed Integer Linear Program (MILP) which is solved using CPLEX. The second approach is a new method that is an extension of a recently introduced MILP, which is solved using a column and row generation approach. In this paper, we benchmark the performance of the methods on networks of two countries (Denmark and The Netherlands). We use the approaches to make daily schedules and we test their real time applicability by performing tests with different disruption scenarios. The computational experiments demonstrate that both models can be used on both networks and are able to find optimal rolling stock circulations in the different planning phases. Furthermore, the results show that both approaches are sufficiently fast to be used in a real-time setting.

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1. Introduction

In passenger railway rolling stock (re)scheduling one of the main goals is to make efficient plans that can accommodate all passengers or, if that is not possible, that minimize the seat shortages. In the planning process, a railway operator tries to match the demand by first selecting an appropriate timetable followed by a matching rolling stock schedule, and finally, by appointing the drivers and conductors to operate the trains in the timetable. We focus in this research on the second stage where a rolling stock schedule must be found given the timetable and passenger demand. Not only the planning process, in which a long time ahead a rolling stock schedule must be determined, is considered, but also the real-time construction of rolling stock schedules due to disruptions.

If during operations an unexpected event causes the timetable, the rolling stock, and crew schedule to become infeasible, then these schedules need to be rescheduled. To ensure that the operations can be resumed quickly, new feasible schedules must be found promptly. The major difference between rolling stock scheduling in the planning phase and rolling stock rescheduling during disruptions is the time available to come up with a solution. Next to that, during the rescheduling phase there is less flexibility since the trains are already running, and choices made before the disruption occurred cannot be

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reversed. A natural consequence is that it may not be possible to assign rolling stock to all train services, i.e., some train services may need to be canceled. Therefore in the rescheduling case it is highly undesirable, but considered feasible, to assign no rolling stock to some services.

There is a body of literature tackling the problem of assigning rolling stock to passenger train services. These papers focus either on the planning process or on the rescheduling process. The majority of the approaches existing in literature considers one specific network, and are not benchmarked against other approaches. In this paper we want to make a start with comparing different approaches, as well as comparing them on different networks. We hope to encourage other researchers to do the same.

We consider two rolling stock (re)scheduling approaches for self-propelled train units. The first approach is based on the algorithm introduced by [Fioole et al. \(2006\)](#). This approach makes use of a general purpose solver (CPLEX) to solve a Mixed Integer Linear Program (MILP) for rolling stock (re)scheduling. The second approach is a new approach, which extends the algorithm introduced by [Haahr et al. \(2014\)](#), and which involves path generation for individual rolling stock units. The method of [Haahr et al. \(2014\)](#) does not consider the order of rolling stock units within a train composition. However, this order is of significant importance to determine which units can be coupled/uncoupled to/from a composition. For example, at certain stations only the unit at the rear of a composition can be decoupled. Therefore we introduce an extension which considers the order of rolling stock units within compositions. This new path based formulation requires adaptations to the column generation heuristic introduced by [Haahr et al. \(2014\)](#). Furthermore, in order to improve the performance in terms of computation time, a row generation variant of the algorithm is introduced as well. The advantage of a path based model in comparison to a flow based model (e.g. the model by [Fioole et al. \(2006\)](#)) is that unit specific constraints such as maintenance required by specific units can be included.

Although the MILPs of the two approaches are different, they are set up such that they solve exactly the same problem. Since both solution methods are exact solution methods, they will come to solutions with the same objective value. Note that the solutions themselves can differ if multiple solutions with the same objective value exist.

We benchmark both approaches on rolling stock scheduling and rescheduling instances of Netherlands Railways (NS) and the Copenhagen Suburban Railway Operator DSB S-tog. In the scheduling instances a timetable and the passenger demand are given. The methods must assign rolling stock compositions to every train service such that the seat coverage of passenger demand is satisfactory while minimizing operational costs. For the rescheduling instances, the original rolling stock schedule, an updated timetable, and passenger demand are given. As the demand will be input, modeling the changed passenger demand during a disruption is considered as out of the scope for this paper. In these rescheduling instances the main objective is to assign an appropriate rolling stock composition to as many train services as possible. However, it is no longer a hard constraint that all train services need a rolling stock composition assigned to them. The secondary objectives are to cover the passenger demand as well as possible and to deviate as little as possible from the original schedule. All deviations from the original schedule require additional shunting movements like couplings and uncouplings of train units. Unplanned or canceled shunting movements require additional communication and coordination with shunting personnel; in some cases even additional shunting personnel must be arranged. Introducing deviations to the planned schedule is thus not preferred, especially since the available time is limited to communicate all changes to the involved crew members.

The remainder of this paper has the following structure. First, the contributions of this paper are discussed in Section 1.1. Thereafter, a literature overview of the existing literature is given in Section 2. The problem description and assumptions are discussed in Section 3. The mathematical formulations for the two solution approaches are discussed in Section 4. In Section 5 computational experiments are presented. The paper is concluded in Section 6.

1.1. Contributions

This paper has several methodological and practical contributions. The methodological contribution is the introduction of a new rolling stock (re)scheduling approach by: (i) extending an existing column generation formulation, and (ii) introducing a new row generation method. The new mathematical formulation is a path based MILP formulation for the rolling stock (re)scheduling problem. In this formulation the order of units within compositions is taken into account. Finally, a row generation method is adopted for a significant speedup in runtime of the solution approach.

The practical contributions are: (i) realistic tests on DSB S-tog and Netherlands Railways instances, (ii) a comparison between different rolling stock (re)scheduling approaches, and (iii) comparisons of different instances within different countries. Note that this is the first work that actually incorporates the order of units within compositions in test instances of DSB S-tog. The transition (i.e. composition) rules have been made in co-operation with DSB S-tog. A comparison is made between different rolling stock (re)scheduling approaches by testing them on the same data sets. These approaches are benchmarked on two different railway networks with cyclic timetables, namely a large train service network in the Netherlands and the suburban network in Copenhagen, Denmark.

2. Literature overview

Two main categories of rolling stock (re)scheduling problems in passenger railways have been studied in literature. The first consists of assigning both carriages and locomotives to trips. Each carriage can be coupled individually and

independently to a convoy (or composition), but at least one locomotive is required to pull the convoy. The other branch of research consists of assigning self-propelled train units that are not required to be pulled by a locomotive. These units consist of a fixed number of carriages and have their own traction engines. It is common that these train units can be coupled together to form larger train compositions.

The problem of assigning locomotives and/or carriages to trains can be applied to passenger trains (see e.g. Brucker et al., 2003; Cordeau et al., 2001; Cordeau et al., 2002; Lingaya et al., 2002), but also to freight trains (see e.g. Ahuja et al., 2005; Bouzaiene-Ayari et al., in press; Fügenschuh et al., 2008; Rouillon et al., 2006; Vaidyanathan et al., 2008). The models used in most of these papers are based on a multi-commodity flow formulation, which are therefore referred to as flow-based models. In these mentioned papers, the approaches are tested only on a single network from one country and are not benchmarked against other approaches.

The problem of (re)scheduling self-propelled units is also considered in multiple publications (see e.g. Alfieri et al., 2006; Borndörfer et al., in press; Cacchiani et al., 2010; Cadarso and Marín, 2011; Cadarso and Marín, 2014; Fioole et al., 2006; Haahr et al., 2014; Nielsen et al., 2012; Peeters and Kroon, 2008). Most of these papers use flow-based approaches. The papers Cacchiani et al. (2010) and Haahr et al. (2014) assign paths (a list of subsequent trips) to individual units. Approaches assigning such paths to individual units are from now on referred to as path-based approaches. In the current path-based approaches the order of units within the composition are not considered. Also for the research on (re)scheduling self-propelled train units it holds that the approaches in the papers are only tested on a single network from one country and that they are not benchmarked against other approaches.

For more references, with respect to papers considering rolling stock (re)scheduling approaches, we refer the interested reader to Cacchiani et al. (2014) and Piu and Speranza (2014).

As explained before, in this paper we consider two rolling stock (re)scheduling approaches for self-propelled train units. The first approach is based on the algorithm introduced by Fioole et al. (2006). This approach makes use of a general purpose solver (CPLEX) to solve a MILP for rolling stock scheduling but could actually be used for rescheduling as well as demonstrated in Nielsen et al. (2012). The second approach is an extension of the algorithm introduced by Haahr et al. (2014) which involves the generation of paths for individual rolling stock units. This algorithm makes use of column generation to solve a MILP. We note that Cacchiani et al. (2010) also investigated a path-based model for the rolling stock scheduling problem. Column generation in combination with rolling stock (re)scheduling is also considered by Peeters and Kroon (2008). However, a different decomposition method was performed that did not involve the generation of paths for individual rolling stock units.

The advantage of the path-based formulation based on Haahr et al. (2014) is that it models rolling stock unit duties explicitly, thereby enabling dealing with unit specific constraints naturally; a good example of such constraints are units requiring a maintenance appointment at a workshop. An optimal solution of the MILP formulation of Fioole et al. (2006) only contains information on which compositions are assigned to which trips and how compositions change between trips. However, it does not produce paths for each individual rolling stock unit. In a post processing step, a simple heuristic can construct these individual routes, because an integer flow can always be decomposed into unit valued path flows (see Ahuja et al., 1993). However, taking constraints on individual units into account is not possible. For a discussion and other solutions for this problem we refer the reader to Wagenaar et al. (2015).

Table 1 gives a summary of the mentioned papers in rolling stock (re)scheduling. Here, we classify every paper, including ours, based on 6 characteristics. Note that we do not compare the papers based on their computation times and practical results, as the approaches of these papers are all tested on different instances with different complexities. Therefore, in this current paper a start is made to gain more insight into the practical value of the approaches by comparing different approaches on the same instances.

From the table it can be concluded that the current path based models (Cacchiani et al., 2010; Cordeau et al., 2002; Haahr et al., 2014) do not take the order of units within a composition into account, while the model in the current paper does not neglect this important issue. Some of the flow based approaches take the order of units within the composition into account (Alfieri et al., 2006; Fioole et al., 2006; Lingaya et al., 2002; Nielsen et al., 2012; Peeters and Kroon, 2008; Wagenaar et al., 2015) or can be adapted to do so (Borndörfer et al., in press). However, as discussed above, the disadvantage of these flow based approaches is that it is not possible to put constraints on the duties of individual rolling stock units.

For a general overview of models for railway rescheduling during disruptions and disturbances we refer the reader to Cacchiani et al. (2014), and for an overview on disruption management processes in general we refer to Jespersen-Groth et al. (2009).

3. Problem description

In this section we describe the rolling stock (re)scheduling problem in more detail, and the assumptions we make.

The rolling stock *scheduling* problem consists of assigning a rolling stock composition to every trip in the timetable of one planning horizon, e.g., one day of operation. The rolling stock *rescheduling* problem consists of assigning rolling stock compositions to as many trips as possible for the remainder of the day after the timetable is adapted due to a disruption. A trip is a part of a train service, as specified by the timetable, between two major stations where the composition of a train can be changed. A trip consists of a departure station, departure time, arrival station, and arrival time. Furthermore, a composition is

Table 1

An overview of existing literature and their contributions. In total 6 characteristics are included: 1. Is the focus on scheduling or rescheduling? 2. Are self propelled units or locomotives and carriages considered? 3. Is the model path or flow based? 4. Is the order of units within a composition taken into account? 5. Is the formulation tested on a single or on multiple networks? 6. Are multiple solution methods compared?

Paper	Scheduling (S) or rescheduling (RS)	Locomotives & carriages (LC) or self-propelled units (SP)	Path or flow based	Order within composition	Single or multiple (national) networks	Benchmark methods
Alfieri et al. (2006)	S	SP	Flow	Yes	Single	No
Borndörfer et al. (in press)	S	SP	Flow	No	Single	No
Brucker et al. (2003)	S	LC	Flow	No	Single	No
Cacchiani et al. (2010)	S	SP	Path	No	Single	No
Cadarso and Marín (2011)	S	SP	Flow	No	Single	No
Cadarso and Marín (2014)	S	SP	Flow	No	Single	No
Cordeau et al. (2001)	S	LC	Flow	No	Single	No
Cordeau et al. (2002)	S	LC	Path	No	Single	No
Fioole et al. (2006)	S	SP	Flow	Yes	Single	No
Haahr et al. (2014)	S	SP	Path	No	Single	No
Lingaya et al. (2002)	S	LC	Flow	Yes	Single	No
Nielsen et al. (2012)	RS	SP	Flow	Yes	Single	No
Peeters and Kroon (2008)	S	SP	Flow	Yes	Single	No
Wagenaar et al. (2015)	RS	SP	Flow	Yes	Single	No
This paper	S + RS	SP	Path	Yes	Multiple	Yes

an ordered set of coupled rolling stock units. The assignment of rolling stock compositions should be done such that it minimizes the number of seat shortages for passengers, the total number of carriage kilometers, and several other objectives. Consequently, the rolling stock (re)scheduling problem essentially is a multi-criteria decision problem.

For each trip a maximum length of the allowed composition is given to ensure that the length of the composition assigned to a trip is not longer than the length of the shortest platform amongst all platforms where the train has a stop. Platform lengths along the network may differ, so different trips may have different maximum composition lengths.

After a trip has been operated, the composition is usually assigned to a next *succeeding* trip. Such a combination of two succeeding trips is also called a *connection* (note that this also holds for end stations of a line). It is allowed to first change the composition before it is used on the next trip. However, the transition from one composition to another must follow certain business rules. Depending on the direction in which the trip is operated and the station layout, it is stated on which side of the composition it is allowed to add (couple) extra rolling stock units and on which side it is allowed to decouple units. Such a composition change is called a *transition*.

The possibility of coupling and decoupling units requires that we keep track of the order of the rolling stock units within the composition. Most of the time it is only allowed to (de)couple on one side of the composition. For example, in the Netherlands, if a train continues in the same direction, coupling is usually done at the front of the train. This will speed up the process since the rolling stock unit could, in this situation, already be placed there before the train arrives. Decoupling will most likely take place at the rear of the train if the train continues in the same direction. As a consequence, the train can leave before the decoupled unit is shunted away. Fig. 1 shows an example of a transition network illustrating how different composition transition rules work. An explanation of the Figure is given in the caption. Keeping track of the order of the units within a composition makes the problem more complicated than a simple multi-commodity flow problem. Furthermore, coupling or decoupling units requires shunting personnel and time, and is therefore penalized in the objective function.

In this paper, we assume that shunting yards associated with certain stations have infinite capacity to accommodate all composition changes and to park decoupled rolling stock units. After decoupling a unit, we require a certain minimum time duration before that unit can be coupled to a new trip in another composition. This restriction reflects the time required to move a decoupled unit to the shunting yard and from the yard back to a platform to be coupled again.

For each trip an estimated passenger demand is given. It is not strictly required that the capacity for passengers of the composition assigned to a trip is equal or larger than the passenger demand for that trip. However, assigning a composition to a trip which cannot accommodate all passengers is penalized.

In contrast, considering depreciation, maintenance and energy efficiency, having more capacity than there is demand for a trip is also not preferred. Therefore, the number of kilometers driven by all carriages is penalized as well. A trade-off between the number of seat-shortages and the number of carriage kilometers must be found in the rolling stock circulation.

We must ensure that the next day the rolling stock schedule can be applied as planned. Therefore, the schedule must be such that at the end of the day the rolling stock units are parked at the shunting yards such that at each shunting yard there are as many rolling stock units as are required at the start of the next day (also called *end-of-day balance*). We allow differences in the end-of-day balance, but against a certain penalty, since for each negative unit difference a new deadheading trip should be planned during the night to re-balance the inventories. Deadheading is expensive in practice since it requires additional manpower and causes additional rolling stock costs.

Especially in the case of rescheduling, it is of significant importance that solutions are found very fast. In case of a disruption, the trains need to keep running, which means that the operator cannot wait for one hour to decide how to adapt the resource schedules in order to handle the ongoing disruption.

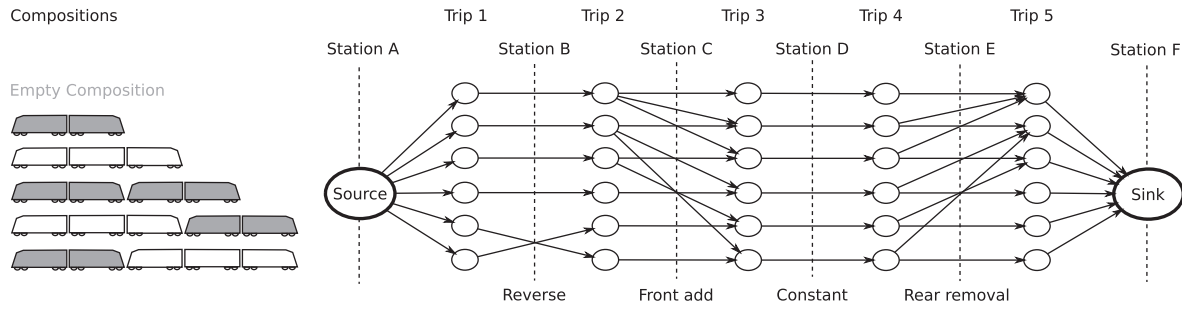


Fig. 1. An example of a transition network illustrating how different rules affect the possible composition transitions from one trip to the successor trip. On the left the example compositions are represented, and to the right the transition network is depicted. Any path from the source to the sink represents a transition scheme that is feasible. A train travels from one station to the next. The transition rules depend on the station layout and business rules. Examples in the figure are: reversing the composition (it returns to the direction it came from), only coupling units to the front, allowing no change, and only allowing units to be decoupled from the rear.

4. Mathematical formulations

In this section we introduce the mathematical formulations of the two solution methods. We present a compact arc-flow formulation first, termed the Composition Model, which is equivalent to the formulation presented by Nielsen et al. (2012) (based on Fioole et al. (2006)). The Composition Model is a state-of-the-art solution approach for solving the considered (re) scheduling problem, and therefore provides a baseline for comparison with other new or extended approaches. The second presented approach is a path based formulation, which is an extension of the model by Haahr et al. (2014). The existing formulation is extended to consider the order of rolling stock units in train compositions in order to prohibit certain infeasible composition changes – similar to the Composition Model. With this addition, the resulting model enables greater expressive power. Rolling stock units can now be subject to unit-specific constraints such as maintenance limitations. A common notation for both methods is presented before both formulations in Sections 4.1 and 4.2 respectively.

In Tables 2–4, the sets, parameters, and variables are introduced and explained respectively.

Both formulations are Mixed Integer Linear Programs, where the objective function is defined as:

$$\text{Minimize : } \sum_{t \in \mathcal{T}} \sum_{c \in \mathcal{C}} \text{compCost}_c^t \cdot y_c^t \quad (1)$$

$$+ \sum_{r \in \mathcal{T}_{\text{con}}} \sum_{c \in \mathcal{C}} \sum_{c' \in \mathcal{C}} \text{transCost}_{c,c'}^r \cdot z_{c,c'}^r \quad (2)$$

$$+ \sum_{u \in \mathcal{U}} \sum_{d \in \mathcal{D}} \text{eodCost} \cdot i_d^u \quad (3)$$

Table 2

List of sets and elements.

Set	Description
$\mathcal{D}$	Set of stations that have associated depot tracks
$\mathcal{C}$	Set of all compositions
$\mathcal{T}$	Set of all trips
$\mathcal{T}_{\text{con}}$	Set of all connections
$\mathbf{s}_r$	The first trip of connection $r \in \mathcal{T}_{\text{con}}$
$\mathbf{t}_r$	The second trip of connection $r \in \mathcal{T}_{\text{con}}$
$\mathbf{d}_r$	The station at which connection $r \in \mathcal{T}_{\text{con}}$ takes place
$\mathcal{U}$	Set of train unit types
$\mathcal{C}_r^2$	Set of all combinations of compositions (c, c') allowed for connection r . This means that composition c is allowed for trip $\mathbf{s}_r$ and composition c' is allowed for trip $\mathbf{t}_r$

Table 3

List of parameters and coefficients.

Parameter	Description
sod_d^u	Starting station inventory of unit type $u \in \mathcal{U}$ at station $d \in \mathcal{D}$
eod_d^u	Target end-of-day inventory of unit type $u \in \mathcal{U}$ at station $d \in \mathcal{D}$
compCost_c^t	Combined costs of trip cancellation, trip seat shortages, and operational costs if composition $c \in \mathcal{C}$ is used on trip $t \in \mathcal{T}$
$\text{transCost}_{c,c'}^r$	Coupling costs of changing from composition $c \in \mathcal{C}$ to $c' \in \mathcal{C}$ between trip $\mathbf{s}_r$ and $\mathbf{t}_r$ of connection r
eodCost	The penalty for a single end-of-day balance shortage

Table 4

List of variables.

Variable	Description
y_c^t	Binary decision variable deciding whether composition $c \in \mathcal{C}$ is used on trip $t \in \mathcal{T}$
$z_{c,c'}^r$	Binary decision variable deciding whether composition $c \in \mathcal{C}$ and $c' \in \mathcal{C}$ are used for trip s_r and t_r respectively
i_d^u	Integer decision variable representing the end-of-day balance shortage for unit type $u \in \mathcal{U}$ at station $d \in \mathcal{D}$

The objective function consists of three parts: costs for assigning compositions to trips, costs for assigning transitions between compositions, and costs for having end-of-day off balances. Trip cancellation, seat shortage and operational costs are included in (1). The shunting costs are included in (2), and the end-of-day shortage costs are accounted for in (3).

In both formulations, the relationship between the composition variables and the transition variables is constrained in order to comply with the physical rules and business logic. For a given trip the rules stipulate which compositions are allowed on the following trip:

$$\sum_{c \in \mathcal{C}} y_c^t = 1 \quad \forall t \in \mathcal{T} \quad (4)$$

$$y_c^{s_r} = \sum_{c' \in \{c_2\} \cup \{c, c_2\} \in \mathcal{C}_r^2} z_{c,c'}^r \quad \forall r \in \mathcal{T}_{\leftrightarrow}, c \in \mathcal{C} \quad (5)$$

$$y_{c'}^{t_r} = \sum_{c \in \{c_1\} \cup \{c_1, c'\} \in \mathcal{C}_r^2} z_{c,c'}^r \quad \forall r \in \mathcal{T}_{\leftrightarrow}, c' \in \mathcal{C} \quad (6)$$

Constraints (4) ensure that exactly one composition is assigned to each trip and Constraints (5) and (6) ensure that a feasible path is found in the transition network, see Fig. 1. Note, that the empty composition is a valid composition, however, this composition assignment has a high penalty, as it corresponds to canceling a trip. The composition assigned to the incoming trip in a connection ($y_c^{s_r}$) must be equal to the actual incoming composition in the chosen transition, as modeled by Constraints (5). Similarly, the composition assigned to the outgoing trip in a connection ($y_{c'}^{t_r}$) must be equal to the actual outgoing composition in the chosen transition, as modeled by Constraints (6).

These constraints do not consider the availability of rolling stock units and do not measure the end-of-day shortages i_d^u . In the following two sections we discuss how the two different models take this into account.

4.1. The composition model

The first option to consider the availability of rolling stock is to keep track of the inventory of rolling stock units at the stations. This option is applied in the formulation of Nielsen et al. (2012), which is based on Fioole et al. (2006). In this section this formulation is summarized.

For this formulation additional parameters and variables are necessary to determine at each station the inventory and the number of coupled and decoupled units. The parameters $coup_{c,c'}^u$ and $uncoup_{c,c'}^u$ indicate how many rolling stock units of type $u \in \mathcal{U}$ should be coupled or uncoupled respectively if the composition changes from $c \in \mathcal{C}$ to $c' \in \mathcal{C}$. These values cannot be negative. For instance, if 2 units of type u need to be coupled during a composition change from composition c to composition c' , then $coup_{c,c'}^u = 2$ and $uncoup_{c,c'}^u = 0$. Furthermore, we assume a certain processing time to shunt a decoupled unit to a shunting yard and to get it back from the shunting yard. Therefore, a parameter τ_r^- is used for each connection r indicating the time it will take before the units that are decoupled during connection $r \in \mathcal{T}_{\leftrightarrow}$ (in between trips s_r and t_r) are available again for coupling. Also a parameter τ_r^+ is required for each connection r indicating the time at which connection r takes place. This is the time the units, which need to be coupled to the composition between trips s_r and t_r , should be available. The station at which connection r takes place is indicated by d_r . Note that this is the station where trip s_r ends and where trip t_r starts.

For connection $r \in \mathcal{T}_{\leftrightarrow}$ and rolling stock unit type $u \in \mathcal{U}$, the non-negative integer decision variables $v_{r,u}^+$ and $v_{r,u}^-$ indicate respectively the number of rolling stock units of type u that are coupled to the composition between trips s_r and t_r , and the number of rolling stock units of type u that are uncoupled from the composition between trips s_r and t_r . Furthermore, a decision variable $inv_{r,u}$, representing the inventory just after connection $r \in \mathcal{T}_{\leftrightarrow}$ (τ_r^+) of rolling stock units of type $u \in \mathcal{U}$ at station d_r , is required.

With these decision variables we can formulate the overall model, where constraints are added to ensure that: (i) the inventory is non-negative at each station and each time period and (ii) the end-of-day off-balance is correctly measured.

$$\text{Objective : (1)–(3)} \quad (7)$$

$$\text{Constraints : (4)–(6)}$$

$$v_{r,u}^+ = \sum_{(c,c') \in \mathcal{C}_r^2} coup_{c,c'}^u \cdot z_{c,c'}^r \quad \forall r \in \mathcal{T}_{\leftrightarrow}, u \in \mathcal{U} \quad (8)$$

$$v_{r,u}^- = \sum_{(c,c') \in \mathcal{C}_r^2} uncoup_{c,c'}^u \cdot z_{c,c'}^r \quad \forall r \in \mathcal{T}_{\leftrightarrow}, u \in \mathcal{U} \quad (9)$$

$$in v_{r,u} = sod_{d_r}^u - \sum_{\substack{r' \in \mathcal{T}_{\leftarrow,}; \\ d_{r'} = d_r; \\ \tau_{r'}^+ \leq \tau_r^+}} v_{r',u}^+ + \sum_{\substack{r' \in \mathcal{T}_{\leftarrow,}; \\ d_{r'} = d_r; \\ \tau_{r'}^- \leq \tau_r^-}} v_{r',u}^- \quad \forall r \in \mathcal{T}_{\leftarrow}, u \in \mathcal{U} \quad (10)$$

$$i_d^u \geq (eod_d^u - sod_d^u) - \sum_{r \in \mathcal{T}_{\leftarrow}, d_r = d} v_{r,u}^- + \sum_{r \in \mathcal{T}_{\rightarrow}, d_r = d} v_{r,u}^+ \quad \forall d \in \mathcal{D}, u \in \mathcal{U} \quad (11)$$

$$i_d^u \in \mathbb{Z}^+ \quad \forall d \in \mathcal{D}, u \in \mathcal{U} \quad (12)$$

$$v_{r,u}^+, v_{r,u}^-, in v_{r,u} \in \mathbb{R}^+ \quad \forall r \in \mathcal{T}_{\leftarrow}, u \in \mathcal{U} \quad (13)$$

Constraints (10) determine the inventory just after a connection takes place. The inventory is equal to the start inventory at the station, minus all units being coupled to compositions in earlier connections, plus all units which are available again after being decoupled from compositions in earlier connections. Note, that since the variables $in v_{r,u}$ are non-negative it also ensures that the inventory is zero or positive at all times. Constraints (11) make sure that the end-of-day shortage is captured. This is equal to the planned difference between the start and end inventory, minus the realized difference between the number of units coupled (departed) and decoupled (arrived) during the day at that station. Constraints (12) and (13) ensure the non-negativity of the number of (de)couplings, inventories and end-of-day shortages. We note that the integrality of the end-of-day balance variables can be relaxed as these will be naturally integer valued.

4.2. The path based model

In Section 4.1 a flow based model for the rolling stock (re)scheduling problem is presented that identifies the correct number of units flowing from station to station. The rolling stock (re)scheduling problem can also be formulated using a path based model. In contrast to the flow based model, the path based formulation explicitly considers the route of each individual rolling stock unit and in this way the availability of rolling stock is guaranteed. The main advantages of this approach is the ability to model unit-specific constraints more easily, e.g., mileage-limitations before next maintenance, or route-choices for units displaying region-dependent commercials. We refer to Haahr et al. (2014) for a study using unit specific constraints. In contrast, the main disadvantage is the added complexity of monitoring the unit paths, in comparison to only considering the flow of unit types in the Composition Model.

The number of possible paths for a rolling stock unit grows exponentially in the number of trips, making the full path model computationally intractable. We therefore propose to solve this method by using column generation, i.e., only a subset of all possible paths is considered. By iteratively solving this reduced model (also known as the master problem) and adding columns with negative reduced cost (found using a so-called subproblem) we are able to get an optimal solution. We refer to Desaulniers et al. (2005) for a detailed introduction to column generation and Branch and Price frameworks.

Recall from Section 1 that this model is an extension of the model presented in Haahr et al. (2014). The model is solved using a Branch and Price framework, i.e., a Branch And Bound (BAB) approach where columns are added dynamically (as needed to prove optimality) at every node in the BAB search tree. The main difference is that we have to consider the order of the units within a composition. For instance, the composition *aba* is different from the composition *baa*. This is important for the composition change rules in practice. The unit in the middle of a composition can, for example, not be uncoupled. Therefore Constraints (4)–(6) are used in this formulation as well.

In the remainder of this formulation, $\mathcal{P}_d^u$ is defined as the set of all paths for unit type $u \in \mathcal{U}$ starting in station $d \in \mathcal{D}$. A path describes a chronological list of trips that are performed by a single rolling stock unit, i.e., a unit's schedule for the planning period. The set of all possible paths is thus denoted by $\mathcal{P} := \bigcup_{u \in \mathcal{U}, d \in \mathcal{D}} \mathcal{P}_d^u$. Shunting operations are implicit as a unit has to be uncoupled (and therefore shunted) whenever the unit exits a trip connection, and coupled whenever the unit, going out of the shunting yard, enters a connection.

The (re)scheduling problem simply consists of assigning exactly one path to each unit, subject to a number of constraints. Therefore, the master formulation contains additional sets of binary decision parameters and variables. The first set of binary parameters, α_p^t , states for each path $p \in \mathcal{P}$ and trip $t \in \mathcal{T}$ whether path p covers trip t ($\alpha_p^t = 1$) or not ($\alpha_p^t = 0$). The second set of binary parameters, β_p^d , takes the value 1 if path $p \in \mathcal{P}$ terminates at station $d \in \mathcal{D}$, and otherwise equals 0. The third set of integer parameters, μ_c^u , indicates how many units of type $u \in \mathcal{U}$ are assigned to composition $c \in \mathcal{C}$. The binary decision variables $\lambda_p \in \{0, 1\}$ determine whether path $p \in \mathcal{P}$ is selected in the final solution or not.

We note that the penalties in Eqs. (1) and (2) of the objective function can be included in the subproblem instead of in the master problem, or alternatively partially included in both. Our preliminary results have shown that it is beneficial for the overall column generation convergence to put a part of the coupling costs in the subproblem. Without awareness of the coupling costs in the subproblem it may generate paths that are unnecessarily expensive, and possibly also incompatible in the master problem.

This leads to the following constraints in the master problem:

$$\text{Objective : (1)–(3)} \quad (14)$$

$$\text{Constraints : (4)–(6)}$$

$$\sum_{d \in D} \sum_{p \in \mathcal{P}_d^u} \alpha_p^t \lambda_p = \sum_{c \in \mathcal{C}} \mu_c^u y_c^t \quad \forall u \in \mathcal{U}, t \in \mathcal{T} \quad (15)$$

$$\sum_{p \in \mathcal{P}_d^u} \lambda_p = \text{sod}_d^u \quad \forall u \in \mathcal{U}, d \in \mathcal{D} \quad (16)$$

$$i_d^u + \sum_{p \in \mathcal{P}_d^u} \beta_p^d \lambda_p \geq \text{eod}_d^u \quad \forall u \in \mathcal{U}, d \in \mathcal{D} \quad (17)$$

$$\lambda_p \in \{0, 1\}, \quad y_c^t \in \{0, 1\}, \quad i_d^u \in \mathbb{Z}_0^+ \quad (18)$$

without Eqs. (4)–(6) the master problem solely consists of finding a set of rolling stock paths. Together with Constraints (15) these constraints ensure that only feasible compositions and composition transitions are made.

In comparison with the Composition Model the path-formulation does not consider inventories at the stations, since it models individual paths for rolling stock units, thereby implicitly handling the inventories. The composition and path variables are linked by Constraints (15) to ensure that the correct number of rolling stock units (and types) are assigned to each trip composition. Constraints (16) ensure that exactly one path is assigned to every rolling stock unit. Note that it is necessary for all units to have a path in order to account for the end-of-day balance. A unit's path contains no trips if the unit stays at a depot for the entire period. The end-of-day balance is enforced in Constraints (17). A slack (which is penalized in the objective) is inflicted if insufficient units terminate at respective stations. Finally, the domains of the variables are shown in (18). We note that the integrality of the end-of-day balance variables can be relaxed as these will be naturally integer valued.

The master problem ((1)–(6), (15)–(18)) is solved iteratively while adding new columns (path variables) that have negative reduced cost. A subproblem is solved, using the duals of (15)–(17), to find such columns or to prove that they do not exist. This problem can be solved as a shortest path problem (or with resource constraints when unit-specific constraints are enforced). When no columns exist with negative reduced cost we have solved the LP relaxation to optimality.

A *Branch and Price* framework is required to solve the problem, as the LP relaxations of the master problem are not necessarily integral. The framework setup and branching rules as described by Haahr et al. (2014) are adopted. Here, branches can be made to stipulate that the number of units on one trip are integral, and that the number of units originating or ending in a station are integral.

In addition to these branching rules, we introduce a new type of branching on the composition variables. Note that for any given trip, an optimal LP solution is not required to assign a value of 1 to one of the composition variables, but can assign a fractional value to some of them as long as the sum is 1. In such cases the branching rule partitions the trip's composition variables into two groups, such that the sum of the fractional values is non-zero in both groups. Note that this is stronger than finding one variable to branch on. Given a trip $t \in \mathcal{T}$ with multiple non-zero composition assignments in an LP relaxed solution, we obtain two branches:

$$\sum_{c \in C_1} y_c^t \geq 1 \quad (19)$$

and

$$\sum_{c \in C_2} y_c^t \geq 1 \quad (20)$$

where C_1 and C_2 define the described partitions. In each branch the sum of one group should be greater than or equal to one. Consequently the sum of the other group is then zero. The same reasoning holds in the other branch.

4.2.1. Subproblem graph example

An example of the subproblem graph is shown in Fig. 2. Note that the underlying graph is acyclic and the weights can be negative. We refer to Haahr et al. (2014) for an in-depth description. Note, that the structure of the subproblem is the same as in Haahr et al. (2014) even though we take the order of train units within a composition into account in the overall method. The example contains three stations whose events (departures and arrivals) are shown as vertically aligned vertices. Stations with associated depot tracks have a paired (gray color) node, representing the depot (i.e. shunting yard). The graph contains one source (S) that has one out-going arc to the first events of all stations, and one target node (T) that has one in-going arc for the last events on different stations. Intermediate arcs represent train movements (moving units between the depot tracks and a platform) and trips traveling from one station to another. Mileage costs are set on trip arcs and coupling costs are set on shunting arcs. Dual costs from the linking Constraints (15) are assigned to the trip arcs. Inventory duals from Constraints (16) are assigned to the arcs extending from the source. Finally, duals from Constraints (17), on the end-of-day balances, are assigned to the target node arcs. We note that, without loss of generality, the subproblem can be changed slightly in implementation as some nodes and edges can be altered or removed.

4.2.2. Alternative formulations

A mixed integer program equivalent to the formulation in (14)–(18) can be obtained by replacing the composition transition rules (Constraints (5), (6)) with Constraints (21) which are based on the composition variables only. For any pair of successive trips say t_1 and t_2 (i.e. a connection) we would require:

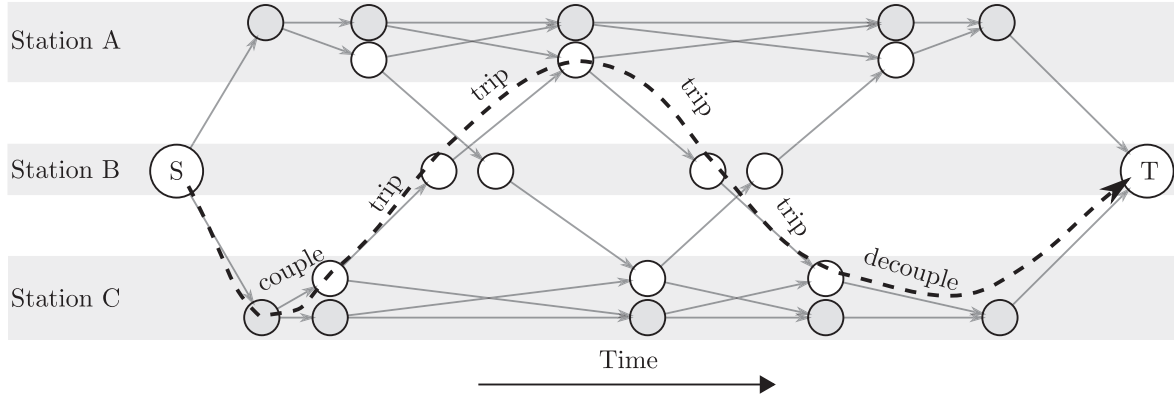


Fig. 2. A simplified example of the acyclic time-space network showing some of the possible coupling arcs. Three stations (A, B and C) are illustrated where only two of them (A and C) have associated depot tracks. An example of a feasible unit schedule has been highlighted using a dashed path.

$$y_c^{t_1} \leq \sum_{c' \in C_c^3} y_{c'}^{t_2} \quad \forall c \in C \quad (21)$$

Here C_c^3 is the set of allowed compositions in t_2 following the composition c on trip t_1 . These constraints limit the origin composition c , for example $y_c^{t_1}$ cannot be set to 1 if the following trip has an incompatible composition. Likewise, a similar set of constraints can be defined that limit the target composition. The resulting formulation has fewer variables and constraints, which may provide a significant speedup when solving the LP relaxations. As a column generation framework relies on solving the LP relaxation many times, this could significantly reduce total computation time. However, preliminary results show that this alternative formulation weakens the LP relaxation too drastically; the relaxed solutions are more fractional. The benefit of faster LP solution times does not outweigh the weaker relaxation in general.

Another equivalent formulation to (14)–(18) can be obtained by modeling the composition transition rules of Constraints (5), (6) by conflicting path constraints as in Eq. (22). Given a full enumeration of all possible paths in the formulation presented in Haahr et al. (2014) (which routes all rolling stock units without any restrictions on train compositions), the composition rules can be enforced by including constraints that prohibit the selection of pairs of conflicting routes:

$$\lambda_{p_1} + \lambda_{p_2} \leq 1 \quad \forall (p_1, p_2) \in \Pi \quad (22)$$

where Π now denotes the set of all conflicting paths. Several important composition rules can be modeled using such pairwise conflict pairs, e.g., only allowing a unit to couple or uncouple on one end of the existing train composition, or disallowing incompatible unit types to be combined on train services.

For example, consider a sequence of trips $\{a, b, c, d\}$ where no intermediate turnings take place. Say path p_1 services all trips, path p_2 only services trips $\{b, c\}$, and path p_3 only services trips $\{b, c, d\}$. Assume that the station between trips a and b only allows units to couple and uncouple at the front of the train, and that the station between trips c and d only allows units to couple or uncouple at the back of the train. As a consequence, paths p_1 and p_2 cannot coexist, as the latter path cannot be coupled to the front of the train and uncoupled from the back when path p_1 is used as well. Thus $(p_1, p_2) \in \Pi$. On the other hand, p_1 and p_2 are both compatible with p_3 .

The obvious disadvantage of this formulation is the potentially large number of resulting conflict constraints. A delayed cut callback routine, that generates the conflicting constraints as they become violated, can be adopted in a Branch-and-Cut method, and possibly remedy this drawback, as only few paths are expected to enter the Simplex basis. However, the non-static row-dimension of the formulation complicates an exact delayed column generation as there is no obvious effective solution approach for the resulting subproblem, where new variables with negative reduced-costs must be found or proven to not exist. Without formally knowing all the possible conflict constraints in advance, it is non-trivial to formulate a subproblem that can account for all necessary dual contributions.

If a full enumeration of all paths is intractable, then a heuristic framework can be adopted that pre-generates an initial set of paths. When all paths are known in advance, the set of pairwise conflicting paths can be deduced and inserted into the formulation. A MIP solver can then be used to solve the resulting formulation. A disadvantage of this approach is the generation of the initial pool of paths. A set of initial paths must be generated, which must contain good and non-conflicting paths. Some of the paths must be able to contain overlapping trips without being in conflict. One approach can include all paths generated while solving the LP-relaxation of the root node in the formulation presented by Haahr et al. (2014). The resulting paths are not proven to be compatible, but this approach requires no special phase for selecting initial paths.

4.3. Delayed transition constraint generation

Preliminary results show that the LP relaxation of the formulation requires a significant amount of pivot operations to solve. This is highly undesirable since we are continuously solving the formulation after adding columns in every iteration of the column generation process. We discovered that solving the linear relaxation of the master problem with the Dual Simplex Method was significantly faster than using the Primal Simplex Method. However, in column generation we unfortunately rely on the Primal Simplex Method as subsequent LP resolves can be hot-started using the old optimal solution after adding new columns.

These observations did however lead us to experiment with the formulation in order to improve convergence or runtime speed of the linear relaxation. Constraints (5) and (6) do not only represent a significant amount of constraints in the presented formulation, but also have a large influence on which columns are feasible during the convergence process. As a result, unnecessary columns are generated making the approach slow. When converging against the optimal linear relaxation, these constraints are unnecessarily restrictive as the intermediate set of states (fractional LP solutions) is not important. By removing these constraints from consideration initially, the advantages are threefold: we achieve a smaller linear program to resolve, we may gain a faster convergence, and finally we get better duals due to the omission of equality constraints. In the method we therefore propose a variant that dynamically adds Constraints (5) and (6) as they become violated once the linear program has converged. This step merely consists of looping through the allowed composition transitions and checking whether the current solution violates any constraint. This extension comes at the cost of more complexity, but preliminary results demonstrated that the benefits outweigh the costs in most instances.

For the sake of simplicity we only add the dynamic constraints once no negative reduced cost columns can be found. Thus, the proposed variant searches for new columns first, and for violated constraints second. Note, after adding new constraints, the framework searches for negative reduced cost columns again since a new constraint changes the solution. The LP model has been solved to optimality once no columns can be generated and none of the constraints is violated.

5. Computational experiments

In this section we present and discuss our computational experiments. We test both models on a scheduling case from Netherlands Railways (NS) and on a scheduling case from the Copenhagen Suburban Railway Operator (DSB S-tog). Additionally, we test the models on different rescheduling cases from both NS and DSB S-tog.

We start with describing the railway network characteristics for both networks. Thereafter, we introduce the parameter settings we have adopted. These parameter settings are chosen after several runs to benchmark their influence. Then, a comparison of the computational results of the scheduling instances is made. Afterwards, the rescheduling instances are described in detail for both the NS and DSB S-tog instances. Finally, we conclude the section with an overview of the different rescheduling results.

5.1. Railway network characteristics

5.1.1. Netherlands Railways

A single instance on the Dutch railway network that spans the major part of the intercity network of NS is selected as computational case. Fig. 3 shows the trajectory of the network. The majority of the lines are in the western part of the

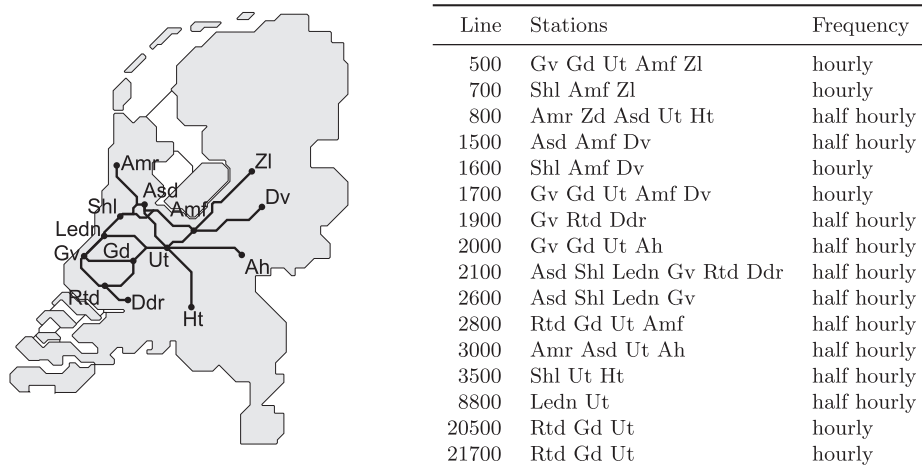


Fig. 3. The NS network considered in the test instances.

Netherlands which is the busiest part of the Dutch railway network. In total we consider the timetable of 16 distinct lines of a certain weekday, namely the Monday. There are different timetables required on the other days. However, the differences between those timetables are small. Therefore, we only consider the timetable of the Monday as scheduling instance. Note that all trips of the 16 lines during the day are taken into account. In general the frequency of the operated lines is half-hourly, but some lines are operated on an hourly basis as can be seen in the table. This leads to a total of 2324 trips between 14 different major stations. The longest trip is between Zl and Amf and has length 67 km, while the shortest trip is between Ledn and Gvc with length 16 km.

We have two different rolling stock types available. A rolling stock unit of type *a* consists of 4 carriages and has a passenger capacity of 405 seats, and a rolling stock unit of type *b* consists of 6 carriages and has a passenger capacity of 597 seats. The maximum composition length for all trips is 14 carriages in the considered network. This leads in total to 11 different compositions that can be appointed to a trip: *a*, *aa*, *aaa*, *b*, *bb*, *ab*, *ba*, *aab*, *aba*, *baa*, and the empty composition meaning that a trip is canceled.

There are many trips between stations where no shunting is allowed. As a consequence, the compositions of those trips are the same as the ones appointed to their predecessor trips. We use a preprocessing step to merge those trips. After preprocessing, the total number of non-reducible trips is 727, i.e., in fact only 727 trip compositions need to be decided as the rest will be fixed due to the composition transition rules. Shunting is allowed at the stations between the remaining trips. It depends on the station whether (un)coupling activities are allowed at the front or the rear side of the incoming train. Furthermore, it is not allowed to both couple and uncouple train units at the same time from the incoming train.

5.1.2. DSB S-tog

The suburban train service network provided by DSB S-tog in Copenhagen (see Fig. 4) is operated using a weekly periodic timetable and an hourly periodic timetable on a daily basis. Weekdays and weekends are operated using slightly different timetables, and an additional set of night trains are operated on Friday and Saturday nights. This leaves a total of four different timetable schedules: Monday (to Thursday), Friday, Saturday and Sunday. Table 5 summarizes the instance specifics. Note that the A, B, C, and F lines are considered in all cases, while the Monday and Friday instances contain more lines than just these 4 lines. As can be seen, the trains are operated with a frequency of either every 10 or every 20 min. The longest trip is between the stations FS and KL with length 55 km, while the shortest trip is between the stations KK and KH with length 3 km.

Two unit types are used to perform all train services. The smaller unit type *e* is 42 meters long and contains 150 passenger seats, the larger unit type *f* is 83 meters and holds 336 passenger seats. Platform lengths vary from station to station in Denmark as well, and the current lines partition the composition lengths into two sets. The first set allows only small compositions *e*, *ee*, and *f*. The second set also allows larger compositions *eee*, *fe*, *ef*, and *ff*.

There are also many trips between stations where no shunting is allowed. The original number of trips and the number of remaining trips after merging are both showed in Table 5. Shunting is allowed between the remaining trips and it depends on the station whether coupling and uncoupling take place at the front or rear side of the incoming train.

5.2. Rolling stock scheduling

In this section we start with describing the parameters used for the scheduling instances for Netherlands Railways and DSB S-tog. Thereafter we present the results of the scheduling instances.

5.2.1. Instance parameters

5.2.1.1. Netherlands Railways. In the scheduling phase of NS the goal is to determine the start and end inventory at every station, and to appoint compositions to trips to fulfill the passenger demand. This is solved in two steps. First a suitable initial fleet distribution is found. Afterwards the found initial inventory setting is used to create the rolling stock circulation.

In the first step, our objective is to create a rolling stock circulation that covers all demand while using the least amount of carriages. To this end, the model determines the number of rolling stock units required at the start of the day at each station to create such a circulation. We set a small penalty on the number of carriage kilometers. No penalty at all is set on the number of shunting operations applied during the day, and no penalty is given for deviations between the start and end inventory.

Table 5

Four timetables operated by DSB S-tog. The columns respectively show the instance names, total number of stops, total number of trips, total number of non-reducible trips (Trips*), weekday, and finally the lines that are running.

Name	Stops	Trips	Trips*	Weekday	Lines
DSBfri	28,719	4558	886	Friday	A,B,Bx,C,E,F&H
DSBsat	20,474	1916	590	Saturday	A,B,C&F
DSBsun	19,919	1871	574	Sunday	A,B,C&F
DSBmon	28,017	4468	868	Monday	A,B,Bx,C,E,F&H



Line	Stations	Frequency	
A	Kj Sol Und Kh Kk Bud Fm	10&20	20
B	Htaa Val Kh Kk Hl Hot	10&20	20
C	Fs Ba Val Kh Kk Hl Kl	10&20	20
E	Kj Sol Und Kh Kk Hl Hot	10&20	20
F	(Kl) Hl Nel	10&20	20
H	Fs Ba Val Kh Kk Bud Fm	20	20

Fig. 4. The DSB S-tog network considered in the test instances. Only key stations are listed in the table. The table columns show line name, station names and the operated frequencies in minutes for Monday–Friday and Saturday–Sunday respectively.

Table 6
Penalty values used for the NS scheduling case.

Coefficient	NS
Cancel	1,000,000
Mileage	1
Seat shortage	0.5
Shunting	1000
End diff	10,000

Table 7
Penalty values used for the DSB S-tog scheduling case.

Coefficient	DSB S-tog
Cancel	1,000,000
Mileage	0.1
Seat shortage	0.5
Shunting	1000
End diff	10,000

Our first step provides us with a start circulation that contains in total 113 units of type *a* and 58 units of type *b*. If this start circulation is used, NS is able to cover the forecasted passenger demand.

In the second step, we use the start inventory created in the first step as fixed input. It is not allowed to deviate from this start inventory and the objective is then to find a rolling stock circulation that minimizes the costs using the objective coefficients as shown in Table 6. These penalty values are commonly used in literature, but we check the influence of changing the Mileage and Seat Shortage penalty ratio on the results in Section 5.2.2.

Here, *Cancel* means the penalty for canceling a trip, *Mileage* denotes the penalty for driving a single carriage over one kilometer, *Seat Shortage* defines the penalty for having one seat shortage per kilometer, *Shunting* refers to the penalty for doing a shunting operation, and, finally, *End diff* denotes the penalty for a negative difference between the preferred end-of-day inventory and the actual end inventory at a station. Results are shown and discussed in Section 5.2.2.

5.2.1.2. DSB S-tog. DSB S-tog provided us with a start inventory. Thus, for S-tog instances it is not necessary to perform the first step of scheduling, that was described for the NS instance. The start inventory contains 31 units of type *e* and 103 units of type *f*.

Furthermore, we use distinct parameter settings for NS and S-tog since the characteristics of the network and train unit fleets are different. The parameter settings for the S-tog network can be found in Table 7. As can be seen, the only difference between the two settings is the mileage penalty. This is, among other factors, because the rolling stock units of DSB S-tog are measured by length in meters in contrast to the number of carriages. The length in meters is substantially larger than the number of carriages.

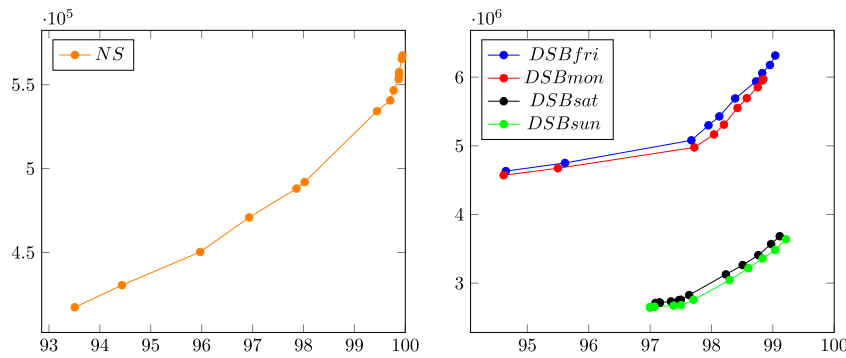


Fig. 5. Trade-off between mileage costs and seat/demand cover for the NS (left) and DSB S-tog (right) instance. The vertical and horizontal axis respectively show the mileage usage (in km) and seat cover percentage.

Table 8

Detailed runtimes using the solution methods on the non-disrupted test instances. The columns respectively show the computation times to solve the instances with the A, B, and N method.

Costs	NS			DSBmon			DSBfri			DSBsat			DSBsun		
	A	B	N	A	B	N	A	B	N	A	B	N	A	B	N
0.1	444	84	27	398	30	14	314	63	14	14	3	2	21	3	3
0.2	545	65	23	575	44	13	423	59	13	21	6	3	29	4	2
0.3	742	274	16	652	345	21	556	48	21	31	6	3	40	4	3
0.4	712	493	17	579	96	14	479	38	12	57	5	4	41	4	4
0.5	709	465	19	653	119	19	664	37	12	58	10	7	35	4	2
0.6	946	781	16	760	47	11	709	40	14	36	8	5	31	4	3

5.2.2. Computational results

In this section we benchmark the two proposed methods for solving the rolling stock scheduling problem on both NS and DSB S-tog instances. We compare both NS and DSB S-tog instances on the scheduling phase. The mathematical formulations are equivalent and we have verified that the optimal objective costs are identical for the two methods. The objective function is a mix of several penalties where the exact balance between them is non-trivial to set. Due to this, there is little justification for solving to true optimality. In practice, it might be enough to accept solutions that are within 1% of optimality, but to be able to compare the approaches accurately we solve to optimality.

In order to justify the parameter values as presented in Tables 6 and 7, we investigated the trade-off between the seat shortage and mileage costs. All other penalties are fixed on their value as given in Tables 6 and 7, only the mileage costs are increased. The results are shown in Fig. 5. As can be seen, there is a steady increase in mileage costs as the seat cover moves towards 100%. However, when we get close to 99% we observe diminishing returns. The final percentages to get a 100% cover are the most difficult percentages. The passenger demand on some trips is too large, as a consequence train units from other stations have to be transported in order to cover the trip with enough passenger demand. Consequently, too large compositions are appointed to the other trips which leads to a solution with larger costs. A cover of 100% is preferred by rolling stock planners, however not at all cost. In practice a seat cover close to 100% with a rolling stock circulation that is not overly expensive is chosen.

The settings of the parameters do not only influence the costs of the rolling stock circulation, but also the computation time to find the circulation. Information about how the computation times are affected by the different penalties for seat shortages are provided in Table 8 for all scheduling instances. Here A denotes the Path Based Model without row generation, B the Path Based Model with row generation, and N the Composition Model. As can be seen, B is able to solve the scheduling instances on average faster than A. This will be discussed in more detail in Section 5.4. Furthermore, N is able to solve the instances even faster. All solution times are acceptable for the planning phase. We assume that a solution method resulting in optimal schedules within 15 min can be used in practice, because there is plenty of time available in the planning phase.

Recall that the Composition Model and the Path Based Model give the same results in terms of objective values. We have selected an appropriate balance in the following benchmarks by choosing the settings as shown in Tables 6 and 7. These settings lead to the results as shown in Table 9. We have left out the results of the A method, because they are all dominated by the B method. Note that in the DSB S-tog cases it is not possible to reach a seat coverage equal to 100%. The seat coverage for all instances is above 98%, so almost all passengers fit in the appointed compositions with a seat. Furthermore, there is no deviation between the start and end-of-day inventories. This is highly appreciated in practice, because otherwise empty trains need to be scheduled to balance the differences. We can conclude that both methods are useful to be used as decision support tool in practice to create rolling stock schedules in both the Netherlands and Denmark.

Table 9

Scheduling results. The columns respectively show the solved instance, the objective value, the mileage costs, the overall seat-shortage costs, the costs for performing shunting operations, the costs for negative differences between start and end inventories, the total seat cover percentage, computation times and columns generated using the *B* method, and finally the computation times for the *N* method.

Instance	Obj	MC	SSC	SC	δ	Cover (%)	<i>B</i> (s)	Columns	<i>N</i> (s)
NS	639,065	553,310	15,755	70,000	0	99.9	465	14,340	19
DSBmon	719,184	555,970	132,214	31,000	0	98.5	119	5937	19
DSBfri	727,159	583,505	119,654	24,000	0	98.6	37	3627	12
DSBsat	418,148	313,469	87,679	17,000	0	98.3	10	1052	7
DSBsun	413,062	297,574	93,489	22,000	0	98.1	4	1266	2

Table 10

Different disruption scenarios NS. The columns show the case numbers, the location of the disruption, the affected time-slot, and the number of trips needing to be rescheduled.

Case	Disrupted area	Time interval	Number of trips
1	Gd-Ut	07:00–10:00	2187
3	Gd-Ut	11:00–15:00	1662
5	Rtd-Gv	11:00–15:00	1662
8	Amf-Ut	11:00–15:00	1662
10	Gv-Ledn	11:00–15:00	1662
12	Asd-Ut	11:00–15:00	1662
2	Gd-Ut	16:00–19:00	1002
4	Rtd-Gv	16:00–19:00	1002
6	Ledn-Ut	16:00–19:00	1002
7	Amf-Ut	16:00–19:00	1002
9	Gv-Ledn	16:00–19:00	1002
11	Asd-Ut	16:00–19:00	1002

5.3. Rolling stock rescheduling

In this section we discuss the computational experiments for the rescheduling instances. For the instances we assume the demand to be the same as in the original situation. See, for instance, Kroon et al. (2015) for a paper that takes dynamic passenger flows into account by means of a simulation step in the rolling stock rescheduling problem. This aspect is outside the scope of the current paper. Note, however, that it is possible to use the approaches as discussed in the framework of Kroon et al. (2015) without any adaptations.

In this section, we first give a brief overview of the different rescheduling cases for Netherlands Railways (Section 5.3.1) and DSB S-tog (Section 5.3.2). Thereafter, the computational results are discussed (Section 5.3.3).

5.3.1. Netherlands Railways

In the following we compare both models on different disruption scenarios from NS. Consequently, we have chosen 12 different scenarios, of which an overview is given in Table 10. The instances are sorted by the complexity of the problem. Note that a problem with many trips that have to be rescheduled is more complex than a problem with little trains that need to be rescheduled. The disruptions take place on the main parts of the network, causing one track to be blocked for at least three hours in both directions. Consequently, there is no railway traffic possible between the stations where the disruption takes place.

The parameter settings used for the NS rescheduling instances are given in Table 11. The largest penalty is still on canceling a trip, this is the most important criteria. The penalties for carriage kilometers and seat-shortage kilometers are the same as in the scheduling case, because in practice there is no time available to determine the optimal balance for every rescheduling instance separately. Three new penalties are used in the rescheduling cases: *Shunting Unplanned*, *Shunting Canceled*, and *End Diff*. *Shunting Unplanned* denotes the cost for doing a shunting operation that was not originally scheduled. Every new (un)coupling action requires shunting crew members to perform them. As a consequence, the shunting crew needs to be rescheduled which costs time and consequently money. So, we want to minimize the number of additional shunting movements. On the other hand, *Shunting Canceled* stands for the penalty for canceling a scheduled shunting movement. Shunting crew members are instructed to do planned shunting movements during the day, canceling such a shunting movement means that the tasks for those crew members have to be canceled. This has to be communicated to the involved crew members. Therefore, we want to minimize the number of canceled shunting operations as well. However, canceling shunting movements is preferred over adding additional shunting movements, because arranging new shunting crew members for a task is usually more difficult than canceling a shunting crew task. Finally, *End Diff* denotes the penalty for having a negative deviation from the scheduled end-of-day balance. This coefficient is kept identical to the one used in the planned instance for the end-of-day deviation.

Table 11
Penalty values used for the NS and S-tog rescheduling cases.

Coefficient	NS	DSB S-tog
Cancel	1,000,000	1,000,000
Mileage	1	0.1
Seat shortage	0.5	0.5
Shunting unplanned	1000	1000
Shunting canceled	100	100
End diff	10,000	10,000

Table 12
Different disruption scenarios DSB S-tog. The columns denote the different case numbers, the location of the disruption, the affected time-slot, and the number of trips needing to be rescheduled.

Case	Disrupted area	Time interval	Number of trips
6	Val-Fs	09:00–10:00	3550
5	Val-Fs	09:00–13:00	3500
12	Kh-Kk	09:00–10:00	3550
11	Kh-Kk	09:00–13:00	3500
3	Val-Fs	11:00–12:00	3000
4	Val-Fs	11:00–15:00	2970
9	Kh-Kk	11:00–12:00	3000
10	Kh-Kk	11:00–15:00	2970
1	Val-Fs	15:00–16:00	1920
2	Val-Fs	15:00–19:00	1870
7	Kh-Kk	15:00–16:00	1920
8	Kh-Kk	15:00–19:00	1870

5.3.2. DSB S-tog

For the DSB S-tog instances, there are also 12 different disruption scenarios, see Table 12. These instances are sorted on their complexity. The disruptions occur either on the Monday or Friday instances, leading to 24 instances in total.

The same objective coefficients are used as in the NS rescheduling instances. The only difference is the mileage penalty, which is 0.1 instead of 1.0, because of the reasons mention in Section 5.2.2.

5.3.3. Computational results

All three approaches give the same results after rescheduling a disruption instance. Fig. 6 gives an overview of the solution costs for the rescheduling results. Here, the objective value, the mileage costs, and the seat-shortage costs are displayed per rescheduled instance. When the difference between objective value and the sum of the mileage and seat-shortage costs is large, then at least one additional trip is canceled during the rescheduling. The other costs are not shown, because they are very small in comparison. The objective value is largest in the cases where trips are canceled. In the other cases, the objective value is close to the mileage costs. So, the mileage costs are much larger than the seat-shortage costs in all cases. This can also be noted by Fig. 7, where the total seat coverage percentage per rescheduling instance is shown. The seat coverage is always between 97% and 99.9%, which means that only few passengers do not get a seat in the train. As a consequence, the penalty for seat-shortages is small. These results are desired in practice, because, next to not canceling trips, keeping a good passenger service is one of the important characteristics during rescheduling.

Besides finding good results, the computation time is of importance as well in real-time. In a disrupted situation, little time is available to reschedule the rolling stock. Therefore, we will compare our models based on their computation times.

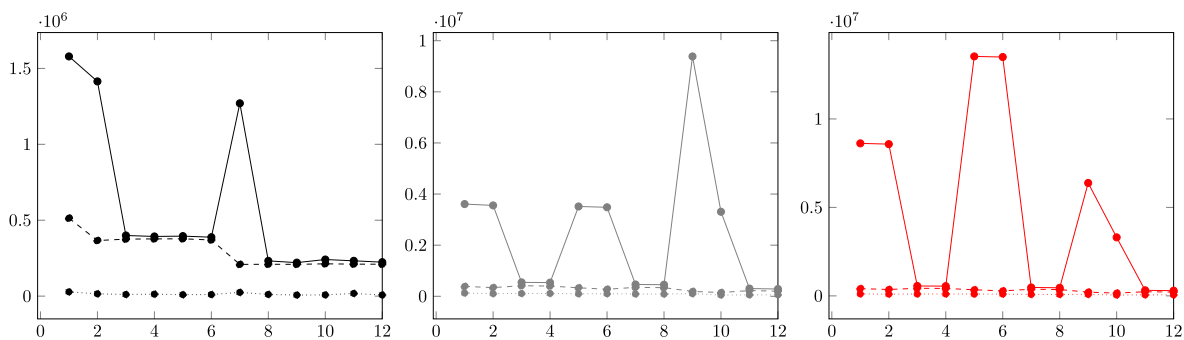


Fig. 6. Results for the NS (left) and DSB S-tog (Monday middle, Friday right) rescheduling instances. A straight line denotes the objective value, a dashed line the mileage costs, and a dotted line the seat shortage costs. The horizontal axis denotes the case numbers and the vertical axis the values.

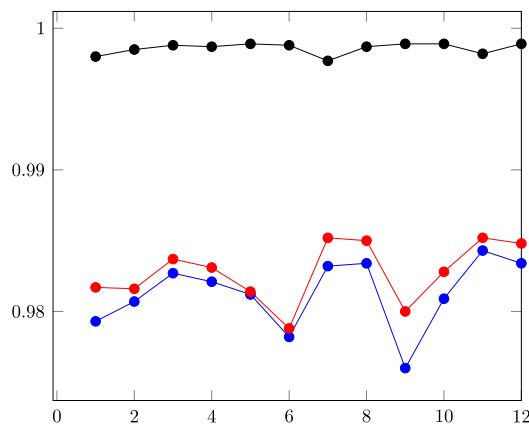


Fig. 7. The seat-coverage for the rescheduling cases. The black line is for the Dutch instances, the blue line for the Friday DSB S-Tog instances, and the red line for the Monday DSB S-Tog instances. The horizontal axis denotes the case numbers and the vertical axis the seat cover percentage, where 1 stands for 100%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 13

Computation time (in seconds) for solving the disruption instances.

Case	NS			DSBmon			DSBfri		
	A	B	N	A	B	N	A	B	N
1	722	269	5	286	207	8	303	40	7
2	138	23	5	176	31	4	151	22	6
3	133	34	3	156	9	3	235	15	6
4	159	39	3	222	17	7	284	15	4
5	157	26	3	72	15	5	113	17	5
6	163	35	3	63	23	5	51	7	3
7	20	6	2	62	6	3	96	10	4
8	11	3	1	106	18	4	103	8	3
9	14	2	1	38	22	4	18	7	3
10	13	13	2	4	2	2	5	2	2
11	10	2	1	8	2	1	13	4	2
12	15	2	1	6	2	2	9	2	2
Average	130	38	3	100	29	4	115	12	4

In Table 13 an overview of all computation times required to solve the different instances with the different models is shown. First of all, the solution method *B* is substantially faster than the solution method *A*. However, both models are considerably slower than method *N*. On the other hand, as explained before, the Path Based Model has other advantages over the Composition Model.

We note that the computation times are influenced by the start time of the disruption – a late disruption involves fewer trips than an earlier disruption, thus a late disruption is easier to solve. We assume that solutions have to be found within 5 min in order to be useful in real-time when a disruption occurs. In conclusion, both methods *B* and *N* are applicable for usage in practice. They are able to produce good results in relatively short time, consequently the rescheduled circulation can be used by practitioners in real-time. On the other hand, too high computation times are observed in some cases for the *A* approach.

5.4. Delayed transition constraint generation

Finally, in this section we present details of the average computation time and average number of columns and rows of the *A* and *B* methods in Table 14. The results show that the dynamic row generation *B* method is on average around 5 times faster than the normal column generation method *A*.

Interestingly, the table shows that, on average, the number of generated columns decreases in *B* for the DSB cases, but increases slightly for the NS cases in comparison with the *A* method. However, we observe a consistent decrease in the number of rows.

Although the *B* method often results in fewer generated columns and rows than the *A* method, there are cases (considering the non-aggregated results) where this is not the case. There exist cases where the runtime of the *B* method is improved in comparison with the *A* method even if there are more columns and rows are generated in total in the *B* method

Table 14

The columns show computation time (in seconds), number of columns and rows generated for solving planning and disruption instances. Each row shows the average over 1 planning case and 12 disruption cases.

Case	A			B			N
	Time	Columns	Rows	Time	Columns	Rows	Time
NS	190	6178	7214	51	6434	3573	2
DSBmon	143	7321	6523	31	4701	5677	4
DSBfri	161	7040	6891	16	4373	4704	4

- we believe this is due to the better convergence property of the *B* method, see Section 4.2. Consequently, we conclude that the *B* method performs better for (re)scheduling the rolling stock when the order of the train units within a composition are of importance.

6. Conclusion

In this paper a comparison is made between two rolling stock (re)scheduling models by testing them on data sets from two countries (the Netherlands and Denmark). The results show that the considered approaches are not limited or tailored to specific networks. Furthermore, it is the first time that the Composition Model is tested on instances of the DSB S-tog network in Denmark and the Path Based Model on instances from Netherlands Railways.

In order to schedule a rolling stock circulation that can be used in practice, a sensitivity analysis of the two models is carried out to determine reasonable values for the mileage and seat-shortage penalties for the train services. The results demonstrate that a higher seat-cover requires significantly more kilometers to be carried out by the carriages. In practice, the operator can decide which ratio between seat-shortages and carriage kilometers suits them best. For the NS instances it was possible to provide a seat for all passengers, but for the DSB S-tog case it was, with the different penalties we have tested, not possible to provide a seat for all passengers. This is due to the fact that the passenger demand is sometimes larger than the capacity of the largest composition we can appoint.

In the current tests the column generation approach of the Path Based Model has longer computation times than the Composition Model. However, the computation times of the path based model are faster than most models from existing literature that are able to take unit specific constraints into account (see for instance Borndörfer et al. (in press), Cacchiani et al. (2010), and Wagenaar et al. (2015)). Therefore, we believe that the column generation approach can be quite interesting, especially when we want to include unit specific constraints, e.g. due to maintenance. Furthermore, we can conclude that adding dynamic row generation to the column generation approach significantly reduced the computation time.

Next to the scheduling instances, we have tested both approaches on rescheduling instances in the Netherlands and Denmark. Disruption instances are smaller than the planning counterparts since all trip decisions up to the start of the disruption are fixed. So in some sense they are easier. However, since we also want to minimize the deviations from the original plan, they are also harder. The experiments show that both models are able to reschedule the rolling circulation fast enough to be used in real-time in both countries.

The proposed approach is of high practical relevance for railway operators. As explained before, in practice there is very limited time available to reschedule the rolling stock during a disruption. Currently, this is still done manually in most of the countries. There are two advantages by using a decision support system that is able to reschedule the rolling stock automatically: (1) The process can be speeded up drastically and (2) The solution quality is most likely much better (i.e. note that our solution approach always provides the optimal solution, while it is doubtful whether the dispatchers are able to find this solution manually).

In future research we want to consider unit specific constraints for the rolling stock, in both a planning and a disruption context. Examples of interesting additional constraints are maintenance appointments and minor rolling stock defects. In the former case units have to reach a maintenance facility in time for an appointment, and in the latter case units with a small defect have to comply with other unit-specific constraints.

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